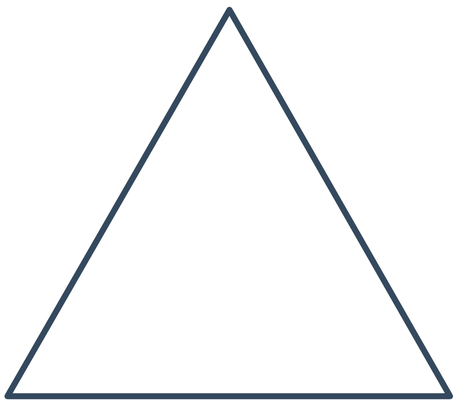
Line symmetry

1 Consider all the single digit numbers, shown below. Which of these numbers have line symmetry?

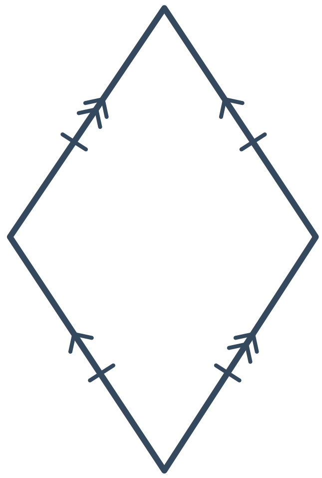
1 2 3 4 5 6 7 8 9 0

2 How many lines of symmetry do the following shapes have?

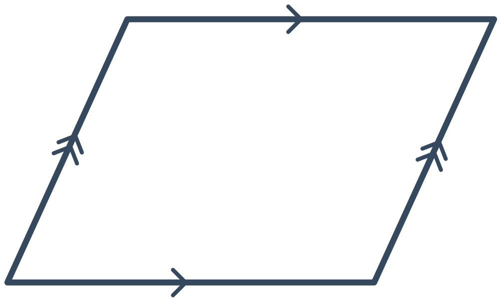
a



b

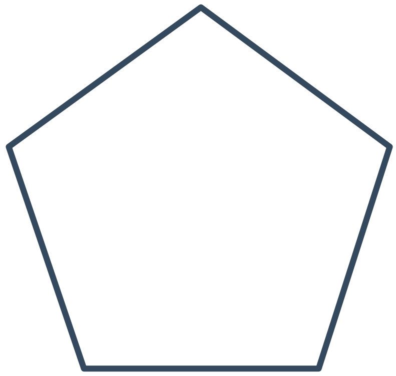


c



d

e



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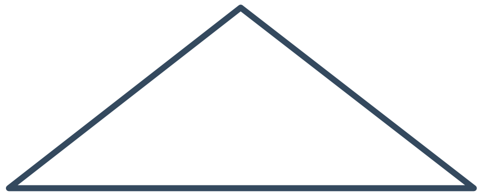
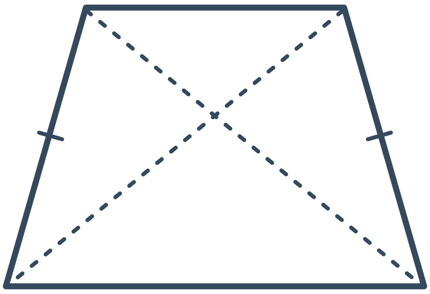
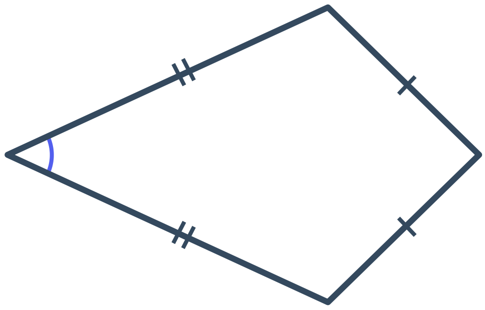


g



h

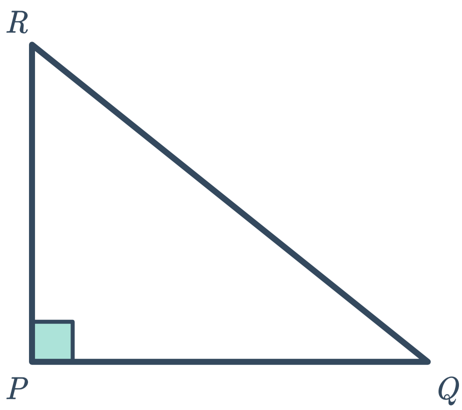


3 Consider the right-angled triangle $\triangle RPQ$ shown. The triangle is reflected across the edge PQ.

Which shape is formed by the triangle and its reflection?

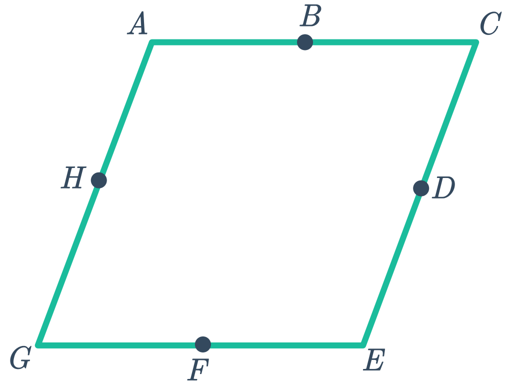
A A square **B** A right-angled triangle

C A rectangle **D** An isosceles triangle



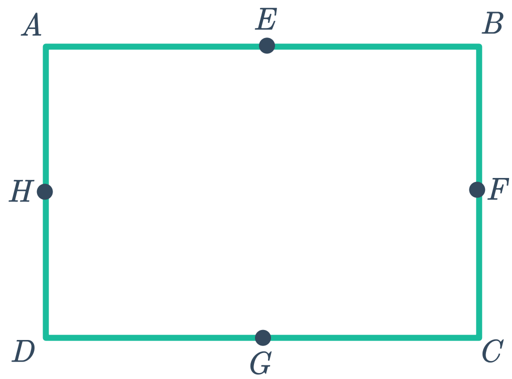
4 Consider the following rhombus. The points B, D, F and H represent the midpoints of the sides of the rhombus.

Write down the pairs of points that can be joined to form a line of symmetry for the rhombus.



5 Consider the following rectangle. The points F, G, H and E represent the midpoints of the sides of the rectangle.

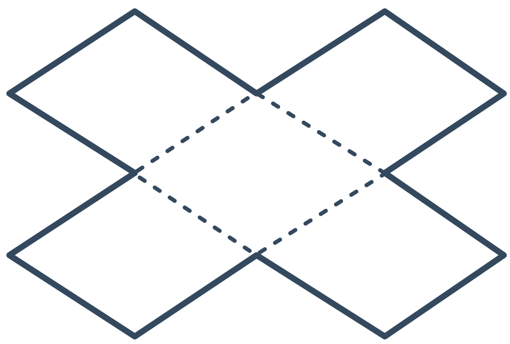
Write down the pairs of points that can be joined to form a line of symmetry for the rectangle.



6 The following shape is constructed from five identical rhombuses:

a How many lines of symmetry does this shape have?

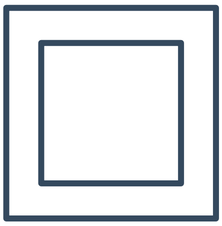
b If each rhombus has a side length of 10 cm, find the perimeter P of the entire shape.



Rotational symmetry

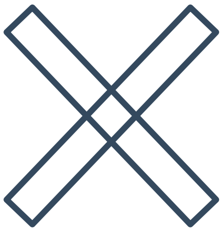
7 Determine whether the following shapes have rotational symmetry:

a

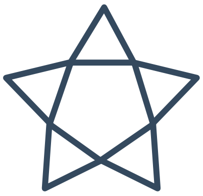


b

c

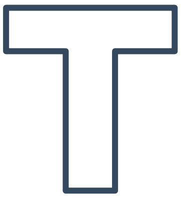


d



8 Determine whether the following figures have rotational symmetry:

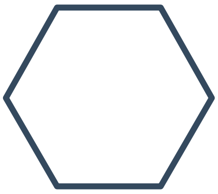
a



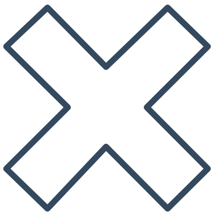
b

c

d



e



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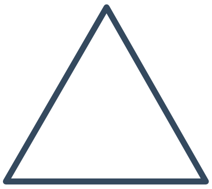
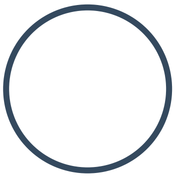


g

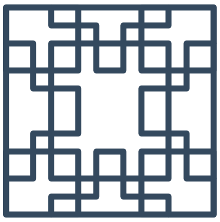
h



i

j

k



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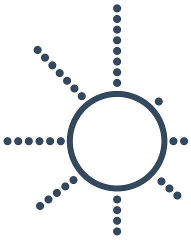

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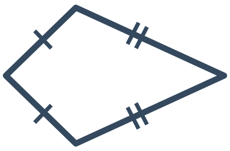

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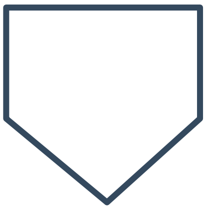
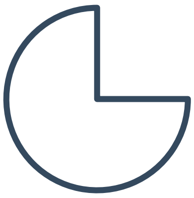
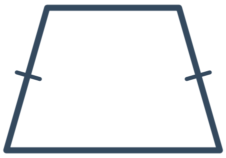


o



p



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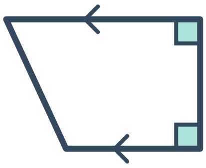

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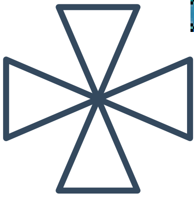

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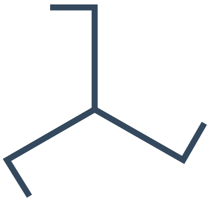

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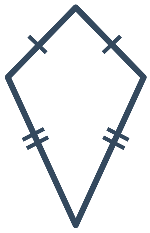

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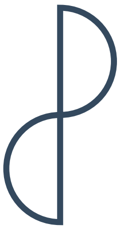

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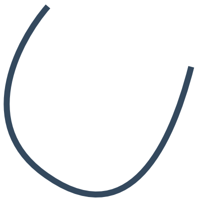

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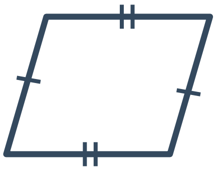

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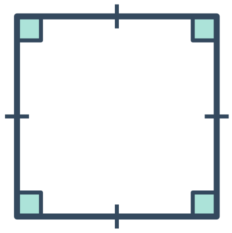
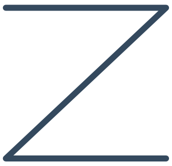


9 Determine whether the following letters have rotational symmetry:

a


b


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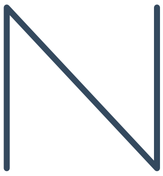

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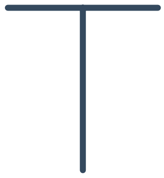
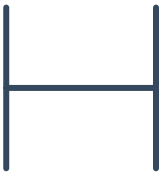
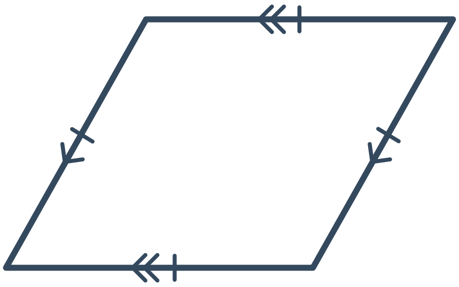


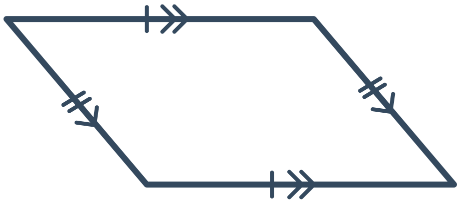
e


10 State the order of rotational symmetry for [redacted]llowing shapes:

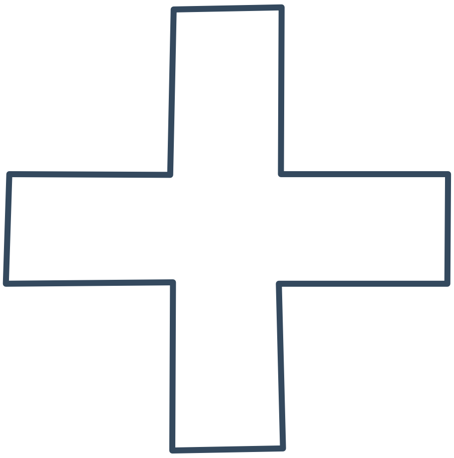
a



b

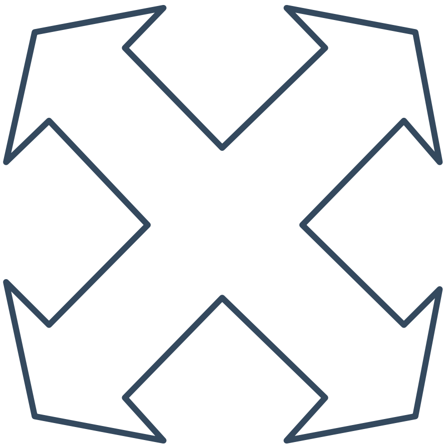


c



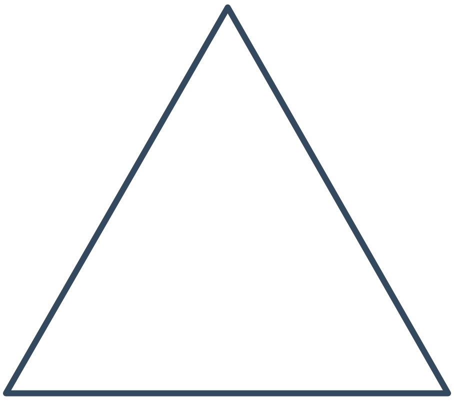
d

e



f

g



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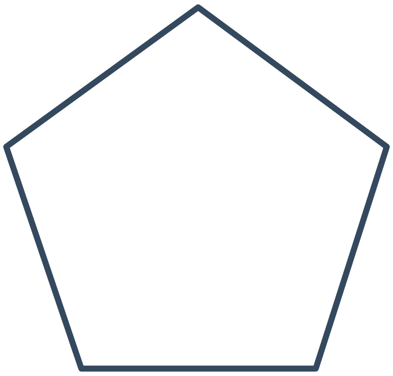
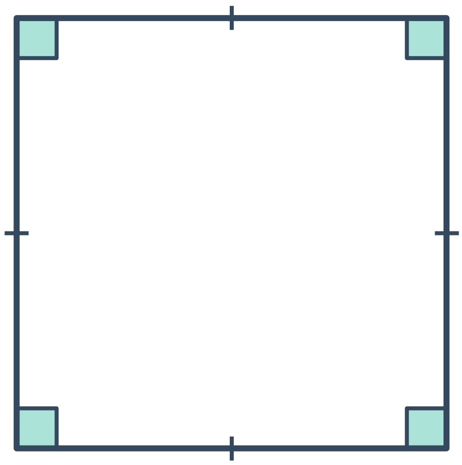


i

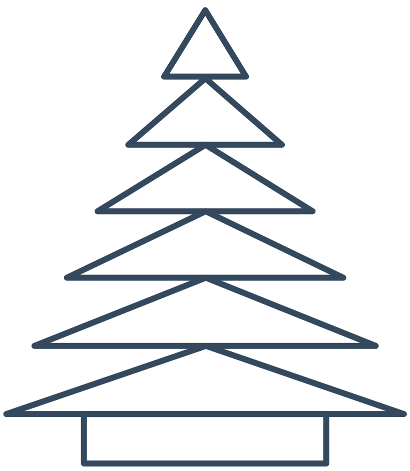
j



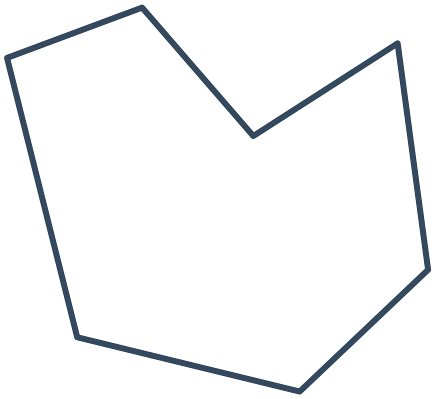
11 State the type(s) of symmetry each of the following figures have:

a

b



c



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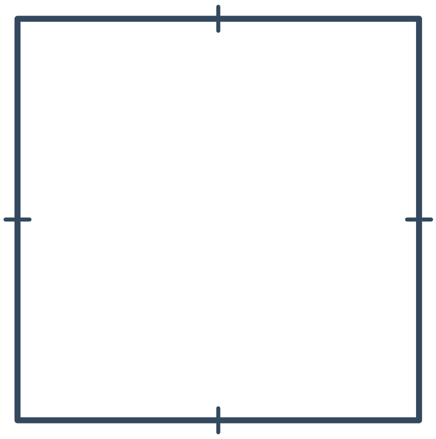

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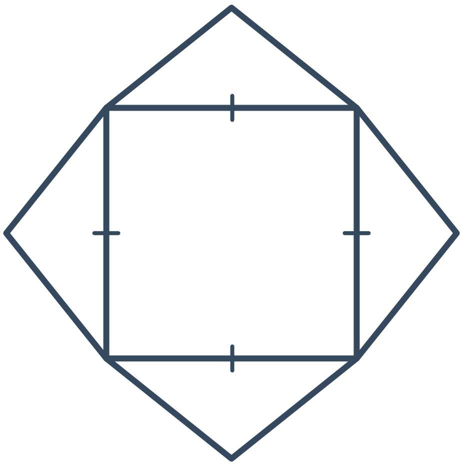

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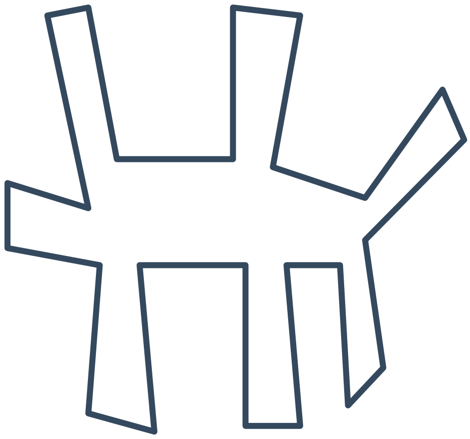

12 If a shape has a rotational symmetry of order 2, state the size of each angle of rotation.

13 If a shape has a rotational symmetry of order 4, state the size of each angle of rotation.

14 Do all isosceles triangles have rotational symmetry?

15 Do all equilateral triangles have rotational symmetry?

Applications

16 Do the following road signs have rotational symmetry?

a

b

17 Consider the road sign shown:

a State the order of rotational symmetry.

b Does the road sign have line symmetry?

18 Dave wants to get a heavy rectangular box from point A to point B by repeatedly turning it over. The width of the box is 30 cm and its length is twice its width.

a The diagram shows the distance travelled after the first and second turn. After 3 turns, what distance along the ground has been covered starting from point A?

b If the distance from point A to point B is 1620 cm, how many times must the box be turned?